Technical Specification

Anisotropic EBL Corrections from Large-Scale Structure

Motivation

Standard EBL models provide an isotropic photon field and corresponding optical depth $\tau_{\text{iso}}(E, z)$. The objective is not to reconstruct the full EBL, but to compute the **line-of-sight-dependent anisotropic correction** induced by observed large-scale structure (LSS).

Scientific goals

1. Quantify anisotropy of the EBL along selected lines of sight:

$$I_{\text{EBL}} = I_{\text{iso}} + \delta I_{\text{aniso}}.$$

2. Compute opacity correction:

$$\tau = \tau_{\text{iso}} + \Delta\tau,$$

   using full angular dependence of $\sigma_{\gamma\gamma}(E, \epsilon, \theta)$.

Data model

- Multi-survey galaxy catalogs (2MRS, 6dF, BOSS DR12).
- Reconstructed LSS overdensity maps ($0.01 < z < 0.18$).
- Coarse radial binning; high angular resolution.
- Maps centered on observer.

Physical assumptions

- EBL baseline is known from homogeneous models.
- Only perturbative contribution from LSS is computed.
- Universe is optically thin in emission: $dI_\nu/ds = j_\nu$.
- High-redshift contribution is effectively isotropic.

Core principle

- Do not reconstruct full EBL.
- Compute only anisotropic correction field δI.
- Treat LSS as perturbation of isotropic background.

Concrete Realization Steps

Anisotropic $\gamma\gamma$ Opacity from LSS Data

Step 1 — Luminosity field construction

Convert galaxy catalogs into a luminosity-weighted field:

$$\delta_L(\mathbf{x}) = \frac{\rho_L(\mathbf{x}) - \bar{\rho}_L(z)}{\bar{\rho}_L(z)}.$$

For each voxel store only available multipoles:

- monopole: ρ_L
- dipole (if statistically stable)
- quadrupole Q_{ij} (main anisotropy carrier)

Higher multipoles are not assumed due to shot noise limitations.

Step 2 — Regime separation

Split Universe into:

- **Local regime ($z < 0.1$):** external catalogs (CosmicFlows4 / Biteau)
- **LSS regime ($0.01 < z < 0.18$):** reconstructed maps
- **Far regime ($z > 0.18$):** isotropic EBL only

Only first two contribute to anisotropy.

Step 3 — Radiation field evaluation

For a point P along the LOS, compute anisotropic intensity:

- Sum contributions of all voxels/cells.
- Use multipole expansion for each cell:

$$I(\hat{n}) \approx \sum_{\ell \leq 2} a_{\ell m} Y_{\ell m}(\hat{n}).$$

- Apply distance weighting and angular kernel.

No full-sky EBL map is constructed.

Step 4 — Numerical acceleration

- Use hierarchical grouping (Barnes–Hut-like), but on **multipole cells**, not galaxies.
- Accept coarse cells with low multipole order at high redshift.
- Preserve explicit local catalog treatment separately.

Step 5 — Optical depth integration

Compute:

$$\Delta\tau(E) = \int dl \int d\epsilon \int d\Omega \, I_{\rm aniso}(\mathbf{x}, \hat{n}) \, (1 - \cos\theta) \, \sigma_{\gamma\gamma}(E, \epsilon, \theta).$$

Then:

$$\tau = \tau_{\rm iso} + \Delta\tau.$$

Step 6 — Output products

- Direction-dependent anisotropy correction $\delta I(\hat{n})$.
- Energy-dependent opacity correction $\Delta\tau(E)$.
- Relative deviation $\Delta\tau/\tau_{\rm iso}$ for each blazar.

Key constraint

No global EBL reconstruction is performed. The method operates entirely on a **hierarchical, multipole-truncated perturbation field** defined separately in local, intermediate, and isotropic regimes.